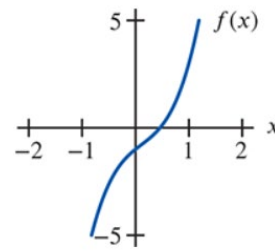


Sec 1.7 – Intro to Limits and Continuity

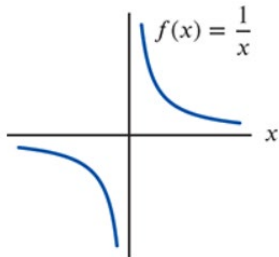
(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

A function is **continuous on an interval** if its graph has no breaks, jumps, or holes in that interval. A function is **continuous at a point** if nearby values of the independent variable give nearby values of the function. Roughly speaking, a function is continuous if it can be drawn without lifting the pencil from the paper.

$f(x) = 3x^3 - x^2 + 2x - 1$ is continuous on any interval:



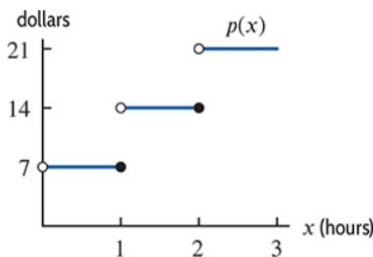
$f(x) = \frac{1}{x}$ is not continuous on any interval containing $x=0$: **(save graph)**



Partner up

Graph the following function:

Example: A company rents cars for \$7 per hour or fraction thereof, so it costs \$7 for a trip of one hour or less, \$14 for a trip between one and two hours, and so on. If $p(x)$ is the price of trip lasting x hours, then its graph (in [Figure 1.101](#)) is a series of steps. This function is not continuous on any open interval containing a positive integer because the graph jumps at these points.



Is it a continuous function? **No, this is a step function, jumps from 7 to 14 with no price in between.**

Note the empty and filled circles. What do they mean?

Considering the continuity of a function at a point is important, because if a function is continuous it means small changes in the independent variable would lead to small changes in the function value. But if a function isn't continuous at a point, a small change in the independent variable could lead to a big change in the function value.

In the car rental example above, 59 min to 1:01 minute would be a \$7 jump in price.

Continuity at a point describes the behavior of a function **near** a point, as well as at the point.

Consider the function $f(x) = \frac{x^2 - 9}{x - 3}$. Is this function defined for all values of x ?

This is an example of a rational function, a numerator divided by a denominator. What is one mathematical rule about division?

Don't divide by 0! Which above happens when $x = 3$.

So even if $f(x)$ is not defined at $x=3$, we can still consider $f(x)$ as it approaches $x=3$.

Definition

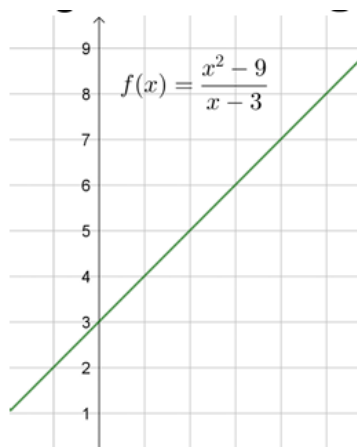
We write $\lim_{x \rightarrow c} f(x) = L$ if the values of $f(x)$ approach L as x approaches c .

We want to find $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$. To match the definition, we are saying that

$f(x) = \frac{x^2 - 9}{x - 3}$. So we want to know what happens to $f(x)$ as x approaches 3.

We can think about this using a **table** or a **graph**:

x	$f(x) = \frac{x^2 - 9}{x - 3}$
2	5
2.9	5.9
2.99	5.99
3.01	6.01
3.1	6.1
4	7

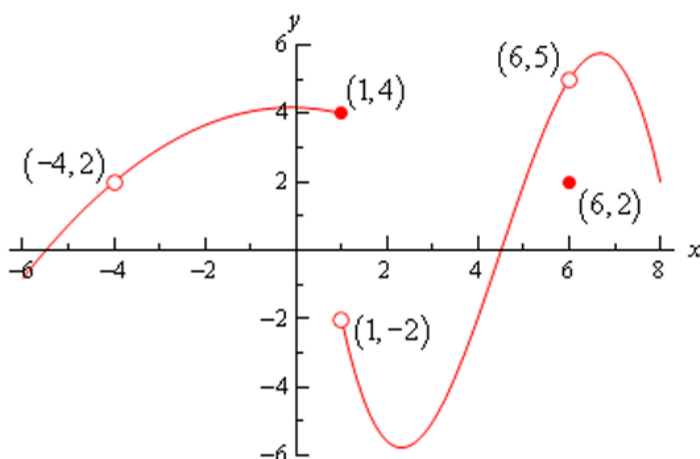


It looks like $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$, even though the function isn't defined at $x=3$.

Let's determine limits from a graph.

Partner up

Let $f(x)$ be as shown in the graph below.



Find the following limits:

- (a) $\lim_{x \rightarrow -4} f(x)$ (b) $\lim_{x \rightarrow 0} f(x)$ (c) $\lim_{x \rightarrow 1} f(x)$ (d) $\lim_{x \rightarrow 6} f(x)$

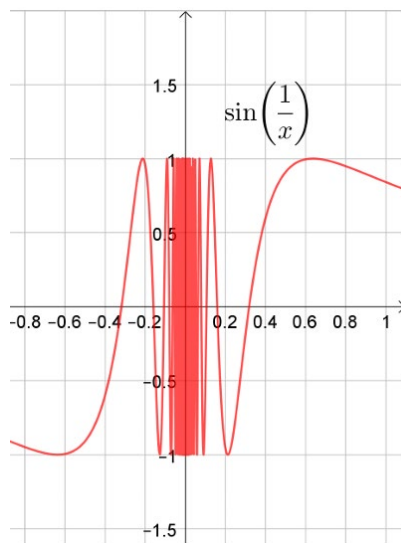
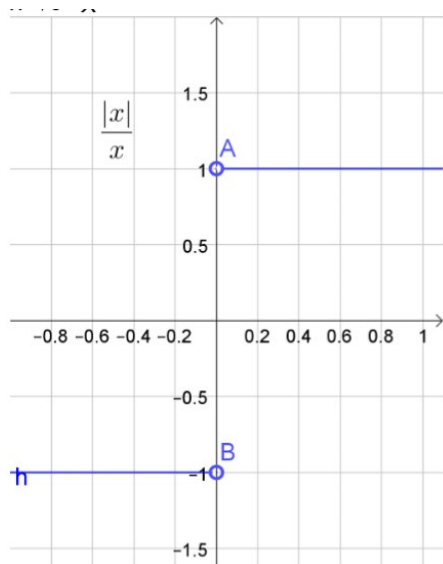
Answers: (a) 2 (b) ≈ 4 (c) does not exist (d) 5

For $x=1$, the function approaches different values from the left, written as $\lim_{x \rightarrow 1^-} f(x)$ and from the right, written as

$$\lim_{x \rightarrow 1^+} f(x)$$

Partner up

Graph the functions $\frac{|x|}{x}$ and $\sin\left(\frac{1}{x}\right)$. Note that we previously graphed $\frac{1}{x}$, which should help as you graph. Use the graphs to determine whether the functions are continuous.



When limits don't exist, we write DNE (Does Not Exist). Typically this is when

- Different values from the two sides, like $\frac{|x|}{x}$
- Infinite values (continuing further and further in the y direction), like $\frac{1}{x}$
- Oscillation, like $\sin\left(\frac{1}{x}\right)$

Definition (Continuity - Formal)

If a function $f(x)$ is continuous at $x = c$, the limit is the value of $f(x)$ there:

$$\lim_{x \rightarrow c} f(x) = f(c).$$

This means that to find the limit of a continuous function at $x = c$, you just have to substitute c into the function!

Now, we have an application of continuity and limits:

Theorem (Theorem 1.1 Intermediate Value Theorem (IVT))

Suppose f is continuous on a closed interval $[a, b]$. If k is any number between $f(a)$ and $f(b)$, then there is at least one number c in $[a, b]$ such that $f(c) = k$.

1. Let f be a continuous function, where $f(-2) = 1$ and $f(3) = -4$. Which values can you guarantee f will take on the interval $(-2, 3)$?

Solution: Since f is continuous, the IVT says that f will take every y -value between $f(-2) = 1$ and $f(3) = -4$ as we move from $x = -2$ to $x = 3$. That is, every y value between -4 and 1 will be taken on the interval $(-2, 3)$

2. **True or False:** $g(x) = x^{100} - 9x^2 + 1$ has a root in $[0, 2]$

Solution: A root is a y -value of 0. Since $x^{100} - 9x^2 + 1$ is a polynomial, it is continuous and we can use the IVT. Since $g(0) = 1$ and $g(1) = -7$ and $-7 \leq 0 \leq 1$, the IVT tells us that there is some x -value c in the interval $[0, 1]$ such that $g(c) = 0$. (Note that $f(2) > 0$, so it wasn't enough to just check the endpoints and see if one is positive and one is negative.)

Explain what is wrong with each statement:

- For any function $f(x)$, if $f(a) = 2$ and $f(b) = 4$, the Intermediate Value Theorem says that f takes on the value 3 for some x between a and b .

Solution: We don't know that the function is continuous, so we can't apply IVT.

- If $f(x)$ is continuous on $0 \leq x \leq 2$ and if $f(0) = 0$ and $f(2) = 10$, the IVT says that $f(1) = 5$.

Solution: The IVT only says that there is *some* x -value c in the interval $[0, 2]$ such that $f(c) = 5$. It doesn't say that the x -value has to be 1.